

Catherine Varas

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Profile

AI & Data Engineer focused on designing and implementing AI systems. I work across the full picture, from the data infrastructure that feeds the models to the orchestration layers that make them work in production. My experience spans computer vision pipelines for scientific research, RAG and automation systems for enterprise clients, and large-scale data architectures across different industries.

Having worked across data engineering, software architecture, and applied AI helps me design systems that are robust and actually hold up in production. I bring a research-driven approach to every problem, I learn the domain, understand the data, and deliver solutions that work reliably at scale. I'm looking for a role in AI, machine learning, or data architecture, in a team tackling meaningful problems.

Industry Experience

Software Engineer — Research Data Pipelines

Mar 2026 – Present

Australian Institute for Bioengineering and Nanotechnology (AIBN) | Brisbane, Australia

- Developed a full-stack data analysis pipeline for fermentation research: volume reconstruction, mass balances, growth rate estimation, and automated phase detection — the same pipeline-thinking needed for scRNA-seq workflows (QC, normalization, clustering, differential expression)
- Built an interactive Dash/Plotly dashboard for researchers to explore data and compare replicates — directly applicable to building visualization tools for single-cell and spatial data exploration
- Documented every formula, assumption, and design decision — the reproducibility standard expected in computational genomics research.

Data & AI Consultant

Jul 2025 – Mar 2026

One51 | Data & Analytics Consultancy | Sydney, Australia

- Designed AI orchestration workflows using Azure Machine Learning, Durable Functions, Azure AI Search, and HuggingFace model serving
- Developed AI RAG systems to automate analytics platform migrations (dashboards and databases=)
- Constructed an AI-based accounting automation pipeline on Databricks integrating OCR, signal normalization, and LLM-assisted semantic resolution
- Built interactive applications for metadata visualization, dependency tracking, and migration gap identification

Associate Software & Data Architect

Jul 2024 – Jan 2025

BBVA | Lima, Peru

- Led migration of millions of contractual records from IBM Mainframe to cloud infrastructure using Apache Spark
- Designed frameworks for data quality, scalability, and compliance with banking regulatory standards
- Developed Java backend routines dynamically querying Mainframe or Oracle systems depending on data recency
- Managed centralized contracts module exposing data across banking systems

Software Architect

Oct 2022 – Jul 2024

BBVA | Lima, Peru

- Engineered scalable batch pipelines using Apache Spark and Java for multi-product Mainframe-to-cloud migration
- Developed reusable Spark routines for data transformation, validation, reconciliation, and loading in Parquet format
- Automated business-rule validations and anomaly detection across millions of contract records
- Deployed and maintained 70+ reusable components (Java backend, Spark batch, database scripts)

BI & Data Analyst

Jul 2021 – Feb 2022

Vida Software SAC | Lima, Peru

- Built Big Data pipelines for performance reporting; automated daily processing of 200+ SFTP files
- Designed dashboards for inventory and sales forecasting for multinational retail clients
- Built integration pipelines between Hasura GraphQL and SQL Server; updated Node.js/Vue.js components

Education

Master in Data Science

2025 – June 2026

University of Queensland | Brisbane, Australia

- Coursework: Cloud Computing, Machine Learning, Statistics, Artificial Intelligence, Deep Learning, Bioinformatics

Master of Artificial Intelligence

2024 - 2026

University of La Rioja | Spain

- Coursework: Generative AI, Machine Learning, Computer Vision, Neural Networks

Bachelor of Industrial Engineering

2019 – 2023

Peruvian University of Applied Sciences | Peru

- Coursework: Thermodynamics, Chemistry, Materials and Manufacturing Technology

Associate’s Degree in Computer Science

2019 – 2022

Institute Cibertec | Peru

- Coursework: SQL, Java, Python, C#, Oracle

Research Experience

Research Volunteer — Computer Vision

Nov 2025 – Present

UQ Queensland Brain Institute | Brisbane, Australia

- Built a multi-model pipeline (YOLO + SAM2) for automated animal tracking — directly transferable to cell segmentation and spatial object detection in imaging omics
- Engineered chunked GPU streaming and HPC parallel execution to process long videos without memory overflow — the same infrastructure pattern required for large-scale scRNA-seq and spatial transcriptomics datasets
- Designed a geometric classifier detecting 6 social interaction types from spatial coordinates — analogous to cell-cell interaction analysis in spatial transcriptomics
- Reconstructed neural network architectures from raw checkpoint weights and debugged precision corruption — relevant to adapting and fine-tuning ML models for genomic applications

Technical Skills

Languages	Python, R, Java, SQL, Bash, C#, JavaScript
ML / AI	YOLO, SAM2, Neural Networks, HuggingFace, RAG, scikit-learn, Deep Learning, Computer Vision, Machine Learning
Genomics & Bio	scRNA-seq Pipelines, Sequencing QC, Alignment, Signal Detection, Bioinformatics Workflows
Data Engineering	Apache Spark, Databricks, ETL Pipelines, Parquet, GraphQL, Control-M
Visualization	Dash/Plotly, Power BI, Streamlit, matplotlib, Publication-ready Figures
Cloud & Infra	Azure (ML, Functions, AI Search, DevOps), AWS, GCP, HPC, GPU Streaming
Databases	Oracle, SQL Server, PostgreSQL, MySQL, Elasticsearch
Practices	Reproducible Research, Git, CI/CD (Jenkins, Azure DevOps), Modular Architecture, Testing

Certifications

- Generative AI Fundamentals — Academy Accreditation
- Microsoft Certified: Fabric Data Engineer Associate (DP700)
- Microsoft Certified: Fabric Analytics Engineer Associate (DP600)
- Microsoft Certified: Power BI Data Analyst Associate (PL300)
- AWS Cloud Practitioner
- Microsoft Azure Fundamentals